Brandon Fluegel, PhD

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SUMMARY

Turns field research and human-performance data into user needs and verifiable design inputs for multimodal, safety-critical systems. Ph.D. from **Old Dominion University**'s Simulation in Healthcare Lab, with graduate research on laparoscopic skills training. Founded the human factors function at **EchoStar**, led new-product definition and set perception-derived latency specifications at **Amazon**, ran use-error analysis on a medical workstation at **NASA**, and coordinated clinical device research at **Harvard Medical School**.

RELEVANT EXPERIENCE

EchoStar
Jun 2025 – Present

Staff Product Researcher (Human Factors)

- Founded the human factors function, reporting to the VP of Product; it is now a standing part of concept reviews across streaming software, hardware, and generative AI products.
- Lead contextual inquiry and rapid iterative prototype testing; built the needs-to-requirements trace: each observed need becomes a user or interface requirement with a tolerance and a verification method.
- Set org-wide, perception-based standards for response time, screen layout, and voice and chat interfaces.
- Lead physical ergonomics for the hardware portfolio: identify use errors in testing and define reach, safety, and anthropometric fit specifications at concept stage.
- Mentor junior designers and researchers; wrote the guides they use to scope studies and write requirements.
- Authored "Principles for Agentic Trust," a framework for human oversight of automation (observability, predictability, safe override); accepted at the ACM Conference on Computer-Supported Cooperative Work and Social Computing (**CSCW 2026**), Industry Perspectives track.

J.S. Held
Nov 2024 – Jun 2025

Managing Scientist, Human Factors

- Led three human factors consulting engagements for major consumer-technology companies.

Amazon, Devices Design Group
Jun 2020 – Nov 2024

UX Researcher (Human Factors)

- **Awarded patent US-12532040-B1** and a 2023 Inventor Award for a control whose action adapts to context.
- Sole human factors researcher in the group; partnered with industrial design, engineering, and product to set requirements for the Echo Show and Echo Hub portfolios.
- Led in-home contextual inquiry for Echo Hub, a new product with no usage data; turned unmet needs into interaction requirements and a design direction built for at-a-glance use, carried into later versions.
- Led a multi-year psychophysics program that replaced arbitrary Alexa latency targets with perception-derived specifications; the Alexa Economics team's model projected $50M in operational value over 15 months.

NASA Langley Research Center
Jun 2018 – Jan 2019

Human Factors Ph.D. Intern

- Ran formative simulated-use evaluation of medical workstations for the Lunar Gateway (planned lunar-orbit station) with astronaut candidates, on a VR prototype simulating microgravity.
- Derived the task analysis and ran a usability FMEA, identifying critical tasks and tracing the highest-severity use errors to the layout of controls.
- Removed those use errors by design rather than by warning or training; no critical input errors were observed in the final formative round, and task time fell 30%.
- Wrote layout and anthropometric requirements to NASA-STD-3001 and MIL-STD-1472; when a haptic cue proved infeasible, delivered the same feedback through visual and auditory channels.

Uber
Jun 2019 – Sep 2019

UX Research Ph.D. Intern

- Ran ride-along contextual inquiry in Brazil with drivers who rent their vehicles, defining unmet needs that shaped one consolidated app for rental management, payments, and rides.

Mercedes-Benz Research & Development
Jun 2017 – Aug 2017

Human Factors Ph.D. Intern

- Ran driving-simulator studies of auditory and tactile alerts during Level 2 automated-driving handovers and wrote the resulting alert-modality guidelines.

Brigham and Women's Hospital / Harvard Medical School
Jun 2014 – Jul 2015

Technical Research Assistant, Clinical Neuroscience Laboratory

- Coordinated clinical research on a transcutaneous vagus nerve stimulation device with psychiatrists and patients with depression; ran fMRI, ECG, and cortisol data collection under IRB and HIPAA requirements.

EDUCATION

Old Dominion University

Ph.D. in Human Factors Psychology 2020

Dissertation: interruptions, distractions, and task resumption during medical data entry.

M.S. in Human Factors Psychology 2017

Thesis: interruptions and distractions during laparoscopic skills training (medical students, box trainer); interruptions caused a resumption lag that distractions did not.

CORE EXPERTISE

Research & Requirements

Contextual Inquiry, Field Observation, Task Analysis, User Needs, Design Inputs, User Interface Requirements, Requirements Traceability, Rapid Iterative Testing

Usability Evaluation & Risk

Formative Usability Testing, Simulated-Use Testing, Usability FMEA (uFMEA), Use Error Analysis, Critical Task Analysis

Human Performance

Psychophysics, Cognitive Load Measurement (fNIRS brain imaging, eye tracking), Multimodal Interfaces (Visual, Haptic, Auditory), Human-Automation Interaction, Anthropometrics & Physical Ergonomics

Standards

NASA-STD-3001 and MIL-STD-1472 (human-systems design standards); familiar with IEC 62366-1, ISO 14971, and FDA human factors guidance